DAY 12 — DATA QUALITY & BUSINESS RULES

Topics Covered:

1. Conditional Formatting
2. Data Validation
3. Business-Rule Checks
4. Exception Highlighting

LEARNING OBJECTIVES

By the end of this lesson, you should be able to:

- Highlight important values automatically using Conditional Formatting.
- Control data entry using Data Validation.
- Create formula-based business-rule checks.
- Identify suspicious records using Exception Highlighting.
- Investigate exceptions before correcting or deleting records.

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1. CONDITIONAL FORMATTING

1.1 What Is Conditional Formatting?

Conditional Formatting is an Excel feature that automatically changes the appearance of cells when specified conditions are met.

It helps a business analyst identify important values without manually checking every row.

Examples:

- Highlight negative sales variance in red.
- Highlight positive variance in green.
- Highlight duplicate transaction IDs.
- Identify overdue orders.
- Highlight unusually high or low values.

1.2 Practical Business Example

Suppose we have the following sales data:

Salesperson	Target	Actual	Variance
Rahul	30,000	35,000	5,000
Priya	30,000	24,000	-6,000
Amit	30,000	30,000	0

The Variance column shows the difference between Actual Sales and Target Sales.

Formula:

```
=C2-B2
```

Assuming Target is in column B and Actual is in column C.

Interpretation:

- Positive variance: Actual sales are above target.
- Negative variance: Actual sales are below target.
- Zero variance: Actual sales equal target.

1.3 How to Apply Conditional Formatting

To highlight negative variance in red:

1. Select the variance cells, such as D2.
2. Go to **Home → Conditional Formatting**.
3. Select **Highlight Cells Rules → Less Than**.
4. Enter `0`.
5. Choose a red formatting style.
6. Click OK.

You can create additional rules:

- Greater Than `0` → Green.
- Equal To `0` → Yellow.

1.4 Business Analyst Application

Conditional Formatting helps analysts quickly identify:

- Underperforming salespeople.
- Negative profit or variance.
- Missing or suspicious values.
- Unusual transactions.
- Records requiring investigation.

Important: Formatting makes a value visible. It does not prove that the value is incorrect.

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2. DATA VALIDATION

2.1 What Is Data Validation?

Data Validation is an Excel feature that controls what users can enter into a cell.

It reduces data-entry errors by restricting entries to approved lists, number ranges, dates, or other conditions.

Examples:

- Allow only approved country names.
- Restrict discount percentages to 0%–30%.
- Allow quantities between 1 and 500.
- Restrict dates to an approved period.
- Limit the length of text.

2.2 Practical Example: Country Dropdown

Suppose your company operates in four countries:

- Kenya
- Nigeria
- India
- Tanzania

You want users to select a country from a dropdown instead of typing it manually.

This helps prevent spelling mistakes such as:

- Knya
- Nigerai
- Indai

2.3 How to Create a Dropdown List

1. Select the cells where the country will be entered, such as B2.
2. Go to **Data → Data Validation**.
3. Under Allow, select **List**.
4. Enter the approved countries in Source:

   ```
   Kenya,Nigeria,India,Tanzania
   ```

5. Ensure **In-cell dropdown** is selected.
6. Click OK.

Depending on your regional Excel settings, the list separator may differ. For a longer list, you can store approved values in a separate range and use that range as the source.

2.4 Common Data Validation Types

Business Requirement	Validation Type
Quantity must be between 1 and 500	Whole Number
Discount must be between 0% and 30%	Decimal
Order date must fall within an approved period	Date
Country must come from approved options	List
Text must not exceed a specified length	Text Length

Percentage example: If percentages are stored as decimal values, 30% is represented by `0.3`. Therefore, a valid discount range is 0 to 0.3.

2.5 Business Analyst Application

Data Validation is useful when:

- Multiple employees enter data into the same workbook.
- Standardized country, branch, or product names are required.
- Incorrect numerical entries could affect reporting.
- You want to reduce errors before they enter the dataset.

Important limitation: Data Validation helps control new entries, but it does not automatically clean existing bad data. Pasted or imported data may also bypass some validation restrictions.

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3. BUSINESS-RULE CHECKS

3.1 What Is a Business-Rule Check?

A business-rule check verifies whether a record follows the organization's agreed business logic.

A value can be numeric and still be invalid or suspicious according to the business rules.

Examples:

- Actual sales cannot be negative.
- Target sales must be greater than zero.
- An order quantity must be positive.
- Achievement percentage must be consistent with Actual Sales and Target Sales.
- A transaction ID must be unique if each row represents one complete transaction.

Business rules must be confirmed with the business before treating a record as incorrect.

3.2 Example: Check Negative Actual Sales

Suppose Actual Sales are stored in cell C2.

The business rule states that Actual Sales cannot be negative.

Formula:

```
=IF(C2<0,"Check","OK")
```

How it works:

- If C2 is negative, Excel returns "Check".
- If C2 is zero or positive, Excel returns "OK".

Example:

Actual Sales	Result
25,000	OK
0	OK
-5,000	Check

This example assumes zero sales are allowed.

Negative sales might sometimes represent refunds, returns, or accounting adjustments, so the business context matters.

3.3 Example: Check Target Sales

Suppose Target Sales are stored in B2.

The business rule requires the target to be greater than zero.

Formula:

```
=IF(B2<=0,"Check target","OK")
```

Interpretation:

- If the target is zero or negative, Excel returns "Check target".
- If the target is positive, Excel returns "OK".

3.4 Example: Calculate Achievement Percentage Safely

Achievement Percentage is calculated as:

Achievement % = Actual Sales ÷ Target Sales

A normal formula would be:

```
=C2/B2
```

However, if B2 is zero, Excel returns a division-by-zero error.

A safer formula is:

```
=IF(B2<=0,"Check target",C2/B2)
```

After entering the formula, format the result cells as Percentage.

This formula:

- Flags zero or negative targets.
- Calculates achievement when the target is positive.
- Avoids division by zero.

3.5 Combined Business-Rule Check

Suppose:

- Target is in B2.
- Actual is in C2.

Use the following formula:

```
=IF(C2<0,"Check negative sales",IF(B2<=0,"Check target","OK"))
```

The formula checks the following conditions in order:

1. If Actual Sales are negative, return "Check negative sales".
2. Otherwise, if Target Sales are zero or negative, return "Check target".
3. Otherwise, return "OK".

You can copy this formula down the column to check multiple records.

Business Analyst Insight: A formula can identify a possible problem, but investigation determines whether the record is genuinely incorrect.

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4. EXCEPTION HIGHLIGHTING

4.1 What Is an Exception?

An exception is a record that violates an agreed business rule or appears unusual enough to require investigation.

Exception highlighting makes these records easier to identify.

Common exceptions include:

- Negative sales when prohibited.
- Missing salesperson names.
- Duplicate transaction IDs.
- Invalid dates.
- Achievement percentages that do not match the underlying figures.
- Unusual transaction amounts.

4.2 Practical Example

Consider the following dataset:

Transaction ID	Salesperson	Actual Sales	Status
TX101	Rahul	25,000	OK
TX102	Priya	-2,000	Check
TX103	Blank	15,000	Check
TX103	Amit	20,000	Check duplicate ID

The flagged records require investigation.

- TX102: Check whether negative sales are allowed.
- The blank salesperson: Investigate whether the name is missing or unavailable.
- TX103: Investigate whether the repeated ID is a genuine duplicate or represents multiple valid transaction lines.

4.3 How to Highlight Exceptions

Suppose the Status column is D.

1. Select the status cells, such as D2.
2. Go to **Home → Conditional Formatting**.
3. Select **Highlight Cells Rules → Text that Contains**.
4. Enter `Check`.
5. Choose a red fill.
6. Click OK.

Excel will highlight cells containing the word "Check".

You can also use formula-based Conditional Formatting to highlight entire rows based on their status.

4.4 Duplicate IDs: Investigate Before Deleting

A repeated transaction ID does not automatically mean a duplicate record.

Before deleting anything, ask:

1. What is the data grain?
2. Does each row represent a complete transaction or a transaction line?
3. Is the transaction ID required to be unique at this level?
4. Could the repeated ID be valid because one transaction contains multiple products?
5. Can the records be verified against the source data?

Example:

If one transaction contains three products, the same transaction ID may appear on three rows. Deleting two rows could destroy valid data.

Rule: Identify the exception first, investigate its cause, and correct it only when the evidence supports the correction.

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5. QUICK FORMULA REFERENCE

Purpose	Excel Formula
Calculate variance	`=C2-B2`
Flag negative Actual Sales	`=IF(C2<0,"Check","OK")`
Flag a zero or negative target	`=IF(B2<=0,"Check target","OK")`
Calculate achievement safely	`=IF(B2<=0,"Check target",C2/B2)`
Combine two business-rule checks	`=IF(C2<0,"Check negative sales",IF(B2<=0,"Check target","OK"))`

Assumptions: Target is in column B, Actual is in column C, and the formulas begin on row 2.

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6. PRACTICE QUESTIONS

Question 1: If Variance = Actual − Target, what does a negative variance mean?

Question 2: Which Data Validation type creates an approved country dropdown?

Question 3: What does the following formula check?

```
=IF(C2<0,"Check","OK")
```

Question 4: Why should you check for a zero or negative target before calculating Achievement Percentage?

Question 5: A transaction ID appears twice. What should you investigate before deleting a record?

ANSWER KEY

Answer 1: Actual Sales are below Target Sales.

Answer 2: List.

Answer 3: It flags negative Actual Sales and returns "OK" otherwise.

Answer 4: Division by zero is undefined, and a non-positive target may violate the business rule.

Answer 5: Check the data grain and whether transaction IDs must be unique. Verify the records against the source before deleting anything.

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7. DAY 12 — FINAL SUMMARY

Today, we learned four important techniques for improving data quality in Excel.

1. Conditional Formatting: Makes important values visible through automatic formatting.

2. Data Validation: Restricts data entry to approved values, ranges, or conditions.

3. Business-Rule Checks: Uses formulas to test records against agreed business logic.

4. Exception Highlighting: Helps analysts identify suspicious records and focus their investigation.

BUSINESS ANALYST WORKFLOW

Step 1: Identify a suspicious or invalid-looking record.

Step 2: Check the data grain and relevant business rules.

Step 3: Investigate the cause using the available evidence.

Step 4: Correct the record only when the evidence supports the correction.

Final Lesson: A flagged record is a signal to investigate, not automatic proof that the data is wrong.

UNSTOPPABLE — DAY 12 COMPLETED